A Brief Analysis of Cloud Computing Infrastructure as a Service (IaaS)

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Abstract:- Cloud computing is a rapidly growing technology in every field. Cloud architecture help and assures to remove expensive and heavy computer hardware maintenance. Cloud architecture introduces an IT environment to create a remote infrastructure which is scalable and measure resources. It is used as a service over the internet. It provides cloud services to the users with full reliability, flexibility, and scalability. In this paper, I will cover all the cloud computing concepts, its services and brief analysis of Cloud Infrastructure as a Service. In cloud computing infrastructure as a Services vendor provide the resources in form of infrastructure user just pay and start their work on it and develop applications or any development project which required heavy system to develop such kind of systems.

Keywords:- *Cloud Computing; Cloud Services; Infrastructure as a Service; IaaS Workloads; IaaS Network;*

I. INTRODUCTION

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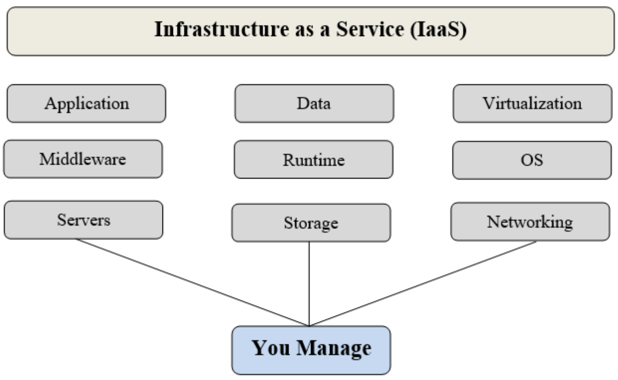

Fig 1: IaaS service network

IaaS model components include Computer Hardware, Network and Internet Connections, Platform Virtualization Cloud Storage, Cloud Software, Utility Computing and Service Level Agreement (SLA). IaaS Communicate with developers to create Virtual private servers, Virtual private storage and Virtual private networks. In IaaS all virtual resources are mapped on the physical system. When the consumer interacts with IaaS service, it requests the resources from the virtual system and these requests are directly forward to the real server which actually works on the user requests.

B. IaaS Security Model

The security model for IaaS is a complete guide for enhancing the security of each level of the IaaS model as shown in the figure – 2 IaaS components of the security model and restriction level. The given figure - 2 front side shows the components – [5]. In this model the three security entities covers the whole components of IaaS. The first entity is the Secure Configure Policy which provides the guarantee of a secure configuration of each component which means each layer in the IaaS either hardware or software is secured. Secured management policy controls the management roles

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of each layer of the security model [5] – [7]. The third and the last entity are the Security Policy Monitoring and Auditing which is responsible for the system life cycle. So all these layers are securing the IaaS security of cloud computing.

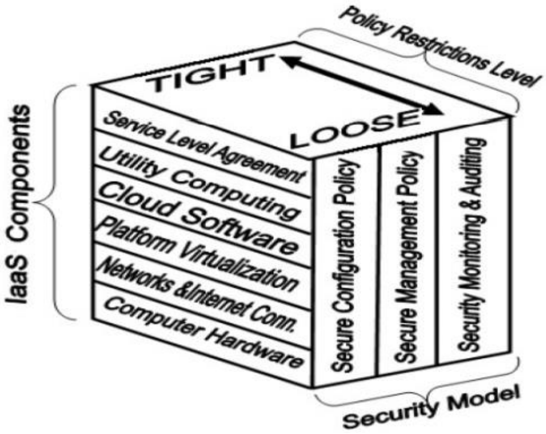

Fig. 2 IaaS Security Model [5]

There are some major security challenges in cloud computing security such as Third Party Handling Data, Cyber Attack, Insider Attack, Government Intrusion, Lack of Standardization, Data Integrity, Lack of transparency and Insecure API's. As the advancement 0f every technology, these challenges are faced and resolved time to time. Cloud Computing provider also focus on all these challenges and resolve these issue.

II. REVIEW OF CLOUD IAAS PROVIDERS

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III. PROPOSED AREA TO IMPROVE IAAS CLOUD SERVICE

In a cloud computing I deeply research about the Cloud Infrastructure as a Service. On the basis of the research, I purposed some solutions the IaaS services to improve some areas of the Cloud IaaS for the better resource provides to the users. The following are the proposed areas of IaaS to improve.

A. IaaS workloads

In IaaS a server is reserved for customer provisioning. Which manage the amount of cloud computing resources according to the user needs for this purpose, users can reserve a server space required for their need which can easily run their workloads. The infrastructure of IaaS runs the servers placed in data centers that offer the services to the cloud users [17] – [18]. The main three layers of the IaaS infrastructure are the RAID storage, virtualized machines, and the interface capacity. These layers are physically systems partitioned as logical units.

B. IaaS Network

In IaaS infrastructure is assigned the private network of user. As like Amazon Elastic Computer Cloud (EC2) behaves every user has its own private separate network not only this you can develop your own private cloud that can be virtualized. Rackspace cloud computing service also follows the Amazon Web Service AWS in the IP assigning model – [19].

C. Pods

Workloads support a specific number of the users. When you reach the limit of the maximum number of virtual machines you need to make a clone for the additional support of the users. A group of them within a particular limit of a cloud computing system is called a pod. The pods are managed by the cloud computing control system (CCS). In AWS, the cloud control system is the AWS management console that manages the Amazon cloud pods [21].

D. Availability Zone.

As we discuss the pods which manage the cloud computing services. Pods also divided into pools within the IaaS region or site called the Availability zone. When a large IaaS infrastructure network pods fail to manage these networks then the zones support the cloud infrastructure to maintain the network and provides the services. In the AWS IaaS infrastructure, the availability zones organized over the company's data centers.

E. Silos

Silos are the cloud computing equivalent of the compute island which is the processing domains which are the outsides of the network infrastructure. When we create a private network within the IaaS framework then the silos help. Silos restrict the users within a certain infrastructure. Silos are responsible for the flexibility of the cloud computing system [19].

With the rapid growth of cloud computing, every organization wants its own private cloud infrastructure which means we improve the cloud computing efficiency to keep maintain and fulfills the customer's requirements. So, here are some major points which can improve cloud efficiency:

- Utilize WANOP and Load balancing
- Optimize your storage environment
- Converged Infrastructure
- Software-defined technologies
- Optimize the end-user experience
- Modernized the end-point
- Monitor and monitor
- Create better security
- Invest in more virtualization
- Automation and orchestration

These above mentioned areas and points are improved, the cloud IaaS service can deliver maximum support to the customers and achieve cloud optimization.

IV. DISCUSSION

A. Cloud IaaS benefits

Cloud infrastructure has several benefits which includes

- Costs of building the new infrastructure resources will reduce by using the shared application resources instead of building a new infrastructure network.
- Management of the single infrastructure is more easy and efficient than many individual network infrastructures.
- To deploy the applications on cloud infrastructure is shorter and easy.
- Cloud infrastructure development is easy to build your own private network in collectively you can use the cloud resources on the pay as you go basis and use as your own private cloud network.

B. Cloud IaaS and 5G

The introduction of 5G in the mobile network enables various numbers of new services that provide shorter latency and higher bandwidth in the network. Cloud-native technologies and edge computing are the cloud infrastructure in 5G [21]. Edge computing provides distributed cloud infrastructure resources close to new business opportunities. Cloud infrastructure is distributed close to the data traffic and consumed more efficient equipment installed.

C. Cloud IaaS transformation philosophy

In this age ultimately everyone needs cloud infrastructure and the 5G accelerates [20]. It ensures the cloud computing service providers to the transformation of cloud infrastructure from the physical network to a cloud environment. The main features of the cloud transformation philosophy includes System Verified, Optimized, Cloud Native and Solution Life Cycle Management.

V. CONCLUSION

Cloud computing IaaS gives an environment where the users can easily and securely build their applications and use cloud computing services based on the user requirement. So with this information, we understand what cloud computing IaaS is. In this paper, I analyze a brief introduction about the cloud computing IaaS, its benefits, and different cloud IaaS providers that provide cloud computing services especially cloud IaaS. Also, I learn what major areas are required for more research on cloud computing IaaS and discuss the cloud computing IaaS components.

Cloud computing is a different platform and have different security challenges because of its open source services which offers the researchers to more explore but I discuss an overview for our topic related things. As we all know that every technology has two affect one is the positive which leads the prosperity and the other is the negative in form of the technology security and challenges. As the same cloud computing also face the various challenges related to its security. But it is no doubt that the cloud computing is the

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most advance and valuable technology of any field which it use.

Although many cloud computing platforms provide the cloud infrastructure as a service for both public and private cloud. This paper is brief details about the IaaS services. The platforms which cover more characteristics of the cloud infrastructure are more efficient.

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